

# Clemente FERRER

*Ph.D. Student in Statistics*

## Research Interests

The interface between Statistics and AI, Extreme Value Theory and Bayesian Nonparametrics.

## Education

2026–Present **Ph.D. in Statistics**, *Pontificia Universidad Católica de Chile*, Santiago, Chile.

2024 **Master of Science in Mathematics**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.

2019–2024 **Mathematical Engineering**, *Universidad Técnica Federico Santa María*, Valparaíso, Chile.

## Articles

### Accepted and Published

1. Vallejos, R., **Ferrer, C.**, Vásquez, A., Díaz, R., Marín, T. (2026) *Modeling first-year engineering student performance during the pandemic*. **The New England Journal of Statistics in Data Science** (Accepted).
2. De Carvalho, M., **Ferrer, C.**, Vallejos, R. (2026). *A Kolmogorov–Arnold Neural Model for Cascading Extremes*. **Extremes** 29, 419–444.
3. Vallejos, R., **Ferrer, C.**, Mateu, J. (2025). *A Concordance Coefficient for Lattice Data: An Application to Poverty Indices in Chile*. **Spatial Statistics** 70, 100936.
4. **Ferrer, C.**, Vallejos, R. (2025). *Is the Effective Sample Size Always Less Than n? A Spatial Regression Approach*. **Statistics and Probability Letters** 218, 110309.

### Submitted

1. Vallejos, R., Osorio, F., **Ferrer, C.** *A new coefficient to measure agreement between two continuous variables*. Submitted to **Biometrical Journal** (Under Review).

### Work in preparation

1. De Carvalho, M., **Ferrer, C.**, Vallejos, R., Ma, T., Sabanis, S. *Generative Cascades of Multivariate Extremes*.
2. Svetlosak, A., De Carvalho, M., He, F., Wu, W., Lee, J., **Ferrer, C.** *KLAN: Kolmogorov-Lorentz-Arnold Neural Networks*.
3. **Ferrer, C.**, Gutiérrez, L., De Carvalho, M., Huser, R., *Amortized Inference for Infinite Mixture Models*.

### Conference Proceedings

1. Carrasco, C., Palme, M., Isalgue, A., **Ferrer, C.** (2026). *Adaptation and Bayesian Calibration of the Eixample Model from Barcelona for Urban Climate in Valparaíso, Chile*. In **Proceedings of the 7th International Conference on Countermeasures to Urban Heat Islands (IC2UHI)**.

## Scholarships and Honors

2026–Present **ANID National Doctoral Scholarship**, *National Agency for Research and Development (ANID).*, Chile
Four-year fully funded fellowship awarded for doctoral studies based on academic excellence.

2025 **Marcos Orrego Puelma Award**, *Chilean Institute of Engineers*, Chile.
Award to the best national engineering graduate of the 2024 cohort.

2024 **Master's Scholarship**, *Universidad Técnica Federico Santa María*, Chile.
Scholarship awarded for outstanding academic performance, consisting of a monthly stipend and tuition fee exemption.

2019–2024 **Academic Excellence List**, *Universidad Técnica Federico Santa María*, Chile.
Award for outstanding academic performance, granted for achieving a grade point average of 80 or higher on a scale of 0 to 100.

2018 **Perfect Score in University Entrance Exam**, *Ministry of Education*, Chile.
Distinction awarded for achieving the highest possible score in the Mathematics section of the Chilean University Selection Test (PSU).

## Research Visits and Schools

Sep 2026 **Edinburgh Summer School on Generative AI for Extremes**, *School of Mathematics, University of Edinburgh.*
Leader of a Computational Session (September 8–September 11, 2026).

Nov 2025 **Applied Mathematics School (AMS)**, *CEMSE, KAUST*, Saudi Arabia.
Fully sponsored one-week program (November 30–December 4, 2025).

Sep 2024 **Research Visit**, *School of Mathematics, University of Edinburgh*, UK.
Hosted by Professor Miguel de Carvalho (September 9–15, 2024).

## Research Projects / Grants

2023–2024 **FONDECYT Project 123001**, *Research Assistant*, Chile.
Agreement measures for spatio-temporal models with applications to image processing.

2023–2024 **Project OEA232**, *Research Assistant*, Universidad Técnica Federico Santa María, Chile.
Quantification of Academic Performance at USM: Its Spatial Distribution and Evolution During the Pandemic.

## Talks

1. *Amortized Inference for Infinite Mixture Models*. VIII Latin American Meeting on Bayesian Statistics & XVIII Brazilian Meeting of Bayesian Statistics, Búzios, Brazil, March 1–5, 2027.
2. *Amortized Inference for Infinite Mixture Models*. XLVIII Jornadas Nacionales de Estadística, Valparaíso, Chile, Octubre 20–23, 2026.
3. *Transformer-Based Approaches for Extremes*. Edinburgh Summer School on Generative AI for Extremes, Edinburgh, UK, September 8–11, 2026.
4. *Generative Cascades of Multivariate Extremes*. 45th Conference on Stochastic Processes and their Applications, Ithaca, US, June 14–20, 2026.
5. *Exploring Transformer-Based Models for Cascading Extremes*. I Workshop on Generative AI Modelling for Extreme Events, Edinburgh, UK, June 13–14, 2025.
6. *Is the Effective Sample Size Always Less Than $n$? A Spatial Regression Approach*. III Congreso Geoestadística y Estadística Espacio-Temporal: Teoría y Aplicaciones, Valparaíso, Chile, November 27–29, 2024.

7. *A Spatial Concordance Coefficient for Areal Data*. XLVII Jornadas Nacionales de Estadística, Valdivia, Chile, November 16–18, 2024.
8. *A Spatial Concordance Coefficient for Areal Data*. II Congreso Geoestadística y Estadística Espacio-Temporal: Teoría y Aplicaciones, Concepción, Chile, December 11–15, 2023.

## Teaching Experience

### Lecturer (*Universidad Técnica Federico Santa María*)

2026 Multivariable Calculus ○ Ordinary Differential Equations.

2025 Regression Analysis ○ Vector Calculus and PDEs ○ Multivariable Calculus ○ Probability Theory ○ Linear Algebra.

### Lecturer (*Universidad Andrés Bello*)

2025 Simulation Systems.

### Teaching Assistant (*Universidad Técnica Federico Santa María*)

2020–2024 Real Analysis ○ Statistical Inference ○ Introduction to Mathematical Engineering ○ Vector Calculus and PDEs ○ Linear Algebra and Integral Calculus ○ Trigonometry and Differential Calculus ○ General Physics III.

## Industry Experience

2026–Present **Chief Risk Officer**, *Quantum Wave*, Chile/US.

2024 **Quant Internship**, *AFP Capital*, Chile.

2023 **Data Science Internship**, *Universidad Técnica Federico Santa María*, Chile.

## Languages

Spanish Native.

English Fluent (IELTS Academic C1, 2025).

## Service

### Reviewed Articles for:

- Journal of Agricultural, Biological and Environmental Statistics.
- Brazilian Journal of Probability and Statistics.